



Flood Polynomial Model

MKM-FPO-001 — Logistic Sigmoid Approximation

MKM Research Labs

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1 Executive Summary

This document presents the mathematical specification, fit quality, and usage of the Flood Polynomial Model (MKM-FPO-001), a closed-form logistic sigmoid with a quadratic polynomial kernel that approximates the per-gauge GBM flood classifier (MKM-FC-001).

The model maps two log-transformed features—water level $\ln(w/s)$ and storm time $\ln((t+1)/T)$ —to $P(\text{flood})$ via six fitted coefficients:

$$P(\text{flood}) = \sigma(a h + b t + c h t + d h^2 + e t^2 + f)$$

where $\sigma(z) = 1/(1 + e^{-z})$ is the standard logistic function.

The model achieves $R^2 = 0.94$, $\text{MAE} = 0.046$, and $\text{RMSE} = 0.099$ against the full GBM probability surface (GAUGE-dc5e88fb, $\text{AUC} = 0.994$, 3.4M samples).

Primary use cases:

- **Single-gauge stress:** Hourly $P(\text{flood})$ estimation in scenario analysis (`scenario.py`), using both water level and storm hour.
- **Portfolio stress:** Time-independent $P(\text{flood})$ for portfolio-level stress P&L (`port_stress.py`), using peak water level only.

Key advantage: No trained `.joblib` model file or scikit-learn dependency required at inference time. The model operates with six fixed coefficients defined in `config/models.py`.

2 Model Purpose and Scope

2.1 Purpose

The Flood Polynomial Model provides a lightweight, closed-form approximation of the GBM flood classifier for two stress testing contexts:

1. **Single-gauge hourly stress:** Given a water level and storm hour, estimate $P(\text{flood})$ without loading a per-gauge trained GBM model.
2. **Portfolio stress:** Provide a time-independent $P(\text{flood})$ for rapid portfolio-level stress P&L calculations where only the peak water level is available.

2.2 Scope

- **In scope:** Closed-form $P(\text{flood})$ estimation from water level and storm hour in log-transformed feature space.
- **Out of scope:** Per-gauge model training, derivative features (velocity, acceleration), multi-gauge spatial correlation.

2.3 Model Chain Position

The polynomial model sits alongside the GBM classifier as a lightweight alternative for stress testing:

$$\text{MKM-FC-001 (GBM surface)} \xrightarrow{\text{curve.fit}} \text{MKM-FPO-001 (polynomial)} \rightarrow \text{Stress P\&L}$$

3 Mathematical Framework

3.1 Model Equation

The model is a logistic sigmoid applied to a quadratic polynomial kernel in two log-transformed features:

$$P(\text{flood}) = \sigma(a h + b t + c h t + d h^2 + e t^2 + f) \quad (1)$$

where:

$$h = \ln\left(\max\left(\frac{w}{s}, 10^{-8}\right)\right) \quad (2)$$

$$t = \ln\left(\frac{\text{hour} + 1}{168}\right) \quad (3)$$

$$\sigma(z) = \frac{1}{1 + e^{-z}} \quad (4)$$

and w is the current water level (m AOD), s is the severe flood warning threshold (m AOD), and $\text{hour} \in [0, 167]$ is the storm hour index.

3.2 Features

Feature	Symbol	Description
Log water level	h	$\ln(w/s)$; water level relative to severe threshold. Zero at severe level, negative below, positive above.
Log storm time	t	$\ln((\text{hour}+1)/168)$; normalised storm time. Ranges from ≈ -5.1 (hour 0) to 0 (hour 167).

Both features are gauge-independent: $h = 0$ when water equals the severe threshold regardless of absolute level; $t = 0$ at the final storm hour regardless of storm duration.

3.3 Polynomial Kernel Terms

The quadratic kernel contains six terms:

Term	Physical Interpretation
$a \cdot h$	Linear water level effect (dominant predictor)
$b \cdot t$	Linear time effect (probability decreases earlier in storm)
$c \cdot h \cdot t$	Water level $\times$ time interaction
$d \cdot h^2$	Quadratic water level curvature (saturation near extremes)
$e \cdot t^2$	Quadratic time curvature (diminishing early-hour uncertainty)
f	Intercept (baseline log-odds)

4 Fitted Coefficients

The six coefficients were obtained by nonlinear least-squares fitting (`scipy.optimize.curve_fit`) of the logistic sigmoid to the GBM probability surface generated from gauge GAUGE-dc5e88fb (AUC = 0.994, 3,393,600 prediction samples).

Symbol	Parameter	Value	Role
a	<code>log_h_s</code>	28.9582	Log water level coefficient
b	<code>log_t</code>	-11.6351	Log storm time coefficient
c	<code>h·t interaction</code>	26.5490	Cross-term coefficient
d	<code>h² quadratic</code>	-22.8893	Water level curvature
e	<code>t² quadratic</code>	-7.0794	Time curvature
f	<code>intercept</code>	-0.8872	Bias term

All coefficients are stored in `config/models.py` as a `FloodPolyCoeffs` named tuple and are fixed at runtime (no per-gauge fitting required).

5 Fit Quality

5.1 Source Data

The polynomial was fitted against the GBM classifier surface for a single representative gauge:

Property	Value
Source gauge	GAUGE-dc5e88fb
Source model	<code>GradientBoostingClassifier</code> (MKM-FC-001)
Source AUC	0.994
Fitting samples	3,393,600
Fit method	<code>scipy.optimize.curve_fit</code> (Levenberg–Marquardt)

5.2 Fit Metrics

Metric	Value
R^2 (coefficient of determination)	0.9395
MAE (mean absolute error)	0.0456
RMSE (root mean squared error)	0.0986

The polynomial captures 94% of the variance in the GBM surface. The model is mildly conservative: it slightly over-predicts $P(\text{flood})$ in the transition zone (alert $\rightarrow$ severe) and slightly under-predicts at the tails. This conservatism is acceptable for stress testing, where false negatives are costlier than false positives.

5.3 Error Distribution

The largest residuals occur in the transition zone where $P(\text{flood})$ changes rapidly from 0 to 1. In the saturated regions ($P \approx 0$ or $P \approx 1$), the polynomial closely matches the GBM predictions.

6 Usage

6.1 Single-Gauge Hourly Stress

For scenario analysis where the full hourly hydrograph is available:

```
1 from src.models.stress.flood_poly import p_flood_poly
2
3 p = p_flood_poly(water_level=5.2, hour=48, severe_level=6.58)
```

This uses both features (h and t) to produce an hour-specific probability estimate.

6.2 Portfolio Stress (Time-Independent)

For portfolio-level stress where only peak water level is available:

```
1 from src.models.stress.flood_poly import p_flood_simple
2
3 p = p_flood_simple(water_level=5.2, severe_level=6.58)
```

The time-independent variant uses the ratio w/s capped at 1.0, providing a simple monotonic mapping suitable for rapid portfolio-wide calculations.

6.3 Hourly Series

For a complete hydrograph:

```
1 from src.models.stress.flood_poly import p_flood_series
2
3 probs = p_flood_series(water_levels=[4.0, 4.5, 5.2, 5.8, 5.5],
4                         severe_level=6.58)
```

7 Implementation Details

7.1 Source Code

- **Primary module:** `src/models/stress/flood_poly.py`
- **Configuration:** `config/models.py` (coefficients, constants)
- **Functions:** `p_flood_poly()`, `p_flood_simple()`, `p_flood_series()`, `model_card()`

7.2 Numerical Safeguards

- **Log floor:** $\ln(\max(w/s, 10^{-8}))$ prevents $\ln(0)$ for zero or negative water levels.
- **Sigmoid clamp:** The exponent z is clamped to $[-30, 30]$ to prevent floating-point overflow in e^{-z} .

7.3 Dependencies

The model requires only the Python standard library (`math` module). No external packages (NumPy, scikit-learn, joblib) are needed at inference time.

8 Sensitivity Analysis

Sensitivity of $P(\text{flood})$ to water level at different storm hours is shown below. All values computed for severe threshold $s = 6.58$ m AOD (GAUGE-dc5e88fb).

Table 1: $P(\text{flood})$ sensitivity to water level and storm hour (severe threshold $s = 6.58$ m AOD).

Water Level (m)	H0	H12	H24	H48	H96	H167
4.00	0.00	0.00	0.00	0.00	0.00	0.00
4.50	0.00	0.00	0.00	0.00	0.01	0.02
5.00	0.00	0.00	0.01	0.02	0.07	0.15
5.50	0.01	0.02	0.05	0.13	0.33	0.53
6.00	0.07	0.14	0.25	0.45	0.71	0.85
6.30	0.20	0.34	0.49	0.69	0.87	0.94
6.58	0.42	0.58	0.71	0.84	0.94	0.97
7.00	0.72	0.82	0.89	0.95	0.98	0.99
7.50	0.93	0.96	0.97	0.99	1.00	1.00

Interpretation:

- At sub-alert levels (4.0–5.0 m), $P(\text{flood}) \approx 0$ regardless of hour, consistent with the GBM surface.
- At the severe threshold (6.58 m), $P(\text{flood})$ ranges from 0.42 at hour 0 (high uncertainty, storm just started) to 0.97 at hour 167 (near certainty, storm almost over).
- Above the severe level (7.0+ m), $P(\text{flood})$ saturates toward 1.0, reflecting overwhelming evidence of flood breach.
- The hour dimension provides meaningful discrimination primarily in the transition zone (5.5–6.6 m), where early-hour uncertainty is largest.

Table 2: Polynomial vs GBM comparison at selected grid points (GAUGE-dc5e88fb, $\Delta w = 0$).

Level (m)	Hour	GBM	Poly	$ \Delta $
5.00	24	0.01	0.01	0.00
5.50	48	0.14	0.13	0.01
6.00	24	0.27	0.25	0.02
6.30	48	0.72	0.69	0.03
6.58	96	0.93	0.94	0.01
7.00	48	0.96	0.95	0.01

The absolute difference between the polynomial and GBM predictions is consistently below 0.05 across the surface, with the largest residuals concentrated in the transition zone around the alert threshold.

9 Model Limitations and Known Weaknesses

9.1 Assumptions

1. **FPO-A1: Single-gauge generalizability.** The six coefficients fitted from GAUGE-dc5e88fb are assumed to be applicable across gauges with different severe thresholds. This is plausible

because the log-transformed feature space normalises water levels, but has not been validated on all 52 Thames gauges.

2. **FPO-A2: Quadratic sufficiency.** A quadratic polynomial in h and t is assumed sufficient to capture the essential shape of the GBM decision surface. The $R^2 = 0.94$ supports this, though a cubic polynomial achieves $R^2 \approx 0.97$ with additional parameters.
3. **FPO-A3: Delta features negligible.** The velocity ($\Delta \ln(w/s)$) and acceleration ($\Delta^2 \ln(w/s)$) features used by the GBM are omitted. This is acceptable because the polynomial is evaluated at steady state ($\Delta w = 0$) in the surface grid, and the GBM attributes only $\approx 6\%$ of feature importance to derivatives.

9.2 Limitations

1. **FPO-L1: Single-gauge fitting.** The model was fitted from one gauge only. Gauges with unusual catchment characteristics (tidal influence, flashy response, regulated flow) may show larger divergence from the polynomial approximation.
2. **FPO-L2: No gauge-specific behaviour.** Unlike the per-gauge GBM classifiers, the polynomial uses the same coefficients for all gauges. Catchment-specific flood dynamics are captured only through the severe threshold normalisation.

10 Change History

Version	Date	Description
1.0	28-Mar-2026	Initial model documentation. Logistic sigmoid with quadratic polynomial kernel fitted from GAUGE-dc5e88fb GBM surface. $R^2 = 0.94$, MAE = 0.046.